

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: B.Tech.

Semester: III

Name of the Course: Advanced Engineering Mathematics

Course Code: TMA-310

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks

Q1		(10 marks)	CO 3													
a)	Perform 10 iterations of Bisection method to find a positive root of $xe^x = 1$ in the interval $[0,1]$.															
OR																
b)	Using Newton-Raphson method find the root of $f(x) = \ln x - x + 3$ in the interval $(4,5)$ correct upto three decimal places.															
Q2		(10 marks)	CO 3													
a)	Find the Lagrange's interpolating polynomial that fits the following data <table><tr><td>x</td><td>-1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$f(x)$</td><td>-1</td><td>11</td><td>31</td><td>69</td></tr></table> Interpolate at $x = 1.5$.			x	-1	2	3	4	$f(x)$	-1	11	31	69			
x	-1	2		3	4											
$f(x)$	-1	11	31	69												
OR																
b)	Construct the forward difference table for the data <table><tr><td>x</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$f(x)$</td><td>-2</td><td>-4</td><td>-4</td><td>-2</td><td>2</td><td>8</td></tr></table> Determine the corresponding interpolating polynomial.		x	-3	-2	-1	0	1	2	$f(x)$	-2	-4	-4	-2	2	8
x	-3	-2	-1	0	1	2										
$f(x)$	-2	-4	-4	-2	2	8										
Q3		(10 marks)	CO 3													
a)	Evaluate $\int_0^1 \frac{dx}{5+3x}$ using Boole's rule.															
OR																

b)	Consider the initial value problem $y' = 3x^2 + \sqrt{y}$, $y(1) = 1$. Using the Euler's method with step size $h = 0.2$, find $y(2)$.	
Q4	(10 marks)	
a)	A random variable X has a CDF of the form $F_X(x) = 1 - \left(\frac{1}{2}\right)^{x+1}, x = 0, 1, 2, \dots$ $= 0, x < 0.$ (i) Find the probability mass function for X. (ii) Find $P(0 < X \leq 8)$.	CO 4
OR		
b)	Four coins are tossed. What is the expectation of number of heads?	
Q5	(10 marks)	
a)	Six dice are thrown 729 times. How many times do you expect atleast three dice to show a 5 or 6.	
OR		CO 4
b)	If X is a normal random variable with mean 20 and standard deviation 5. Find: (i) $P[X - 20 > 5]$ (ii) $P[X \geq 35]$ Use $\phi(1) = 0.8413$ and $\phi(3) = 0.9987$.	